

An opportunity brief. Market, growth paths, and the agentic-AI question.

Prepared by Bruce Longley · TSI-Citadel

For Alan Weiss · Chief Financial Officer · CS Analytical Laboratory

May 2026 · bruce@tsicitadel.ai

Why we're sending this

Three industry shifts compounded in one six-month window.

USP <382> · Dec 1 2025

Elastomer functional suitability shifts to the drug manufacturer. USP: 'early adoption is encouraged.'

FDA agentic AI · Dec 1 2025

All FDA employees — including inspectors — receive secure agentic AI. Regulator is no longer agentially naive.

Nelson acquisitive · May 2026

Sotera Health Q1 2026: 'exploring strategic acquisitions to enhance Nelson Labs' pharmaceutical capabilities.' NJ precedent exists.

CS Analytical at a glance — outside view

The only cGMP-built,
FDA-regulated lab in the world
dedicated to container testing.

Where & what

Clifton, NJ. Doubling lab space, May 2026. USP <1207> deterministic portfolio plus USP <382>, <87>, <788>, gas, micro.

Who — leadership

Brian Mulhall, CEO.
Alan Weiss, CFO, leads finance, BD, HR. Brandon Zurawlow, CSO — PDA TR-86 contributor.

Recent moves

RM Analytical (Feb 2026).
Clifton doubling (May 2026).
Interphex 2026 showcase: IV bag CCIT, <382>, cell-therapy distribution.

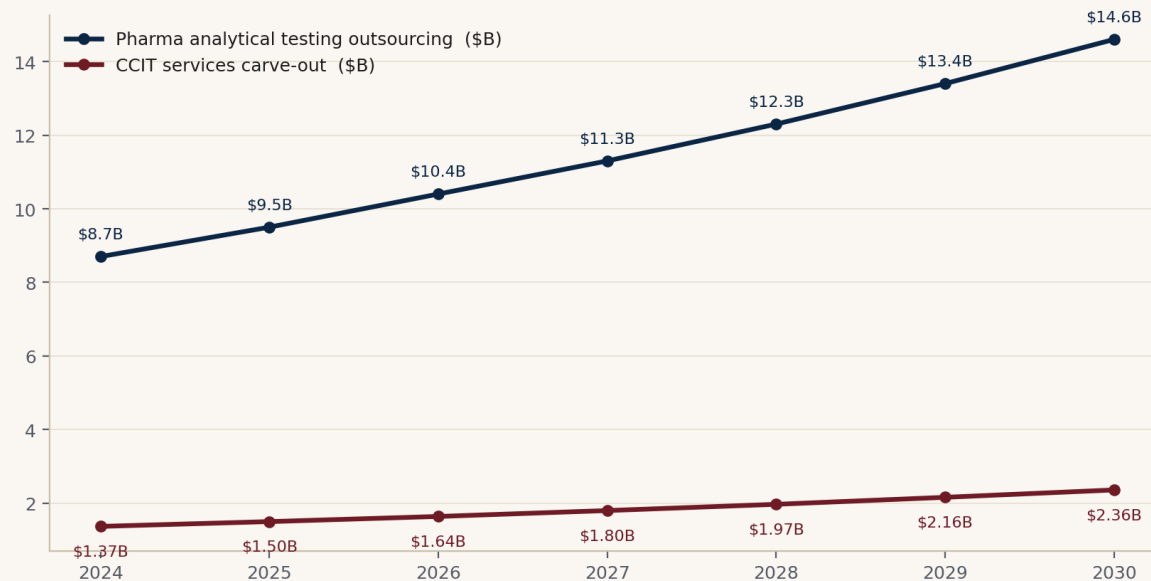
Position

Vendor-neutral. Method-led. Founder-accountable on every study. Inspected by design, not retrofit.

What sits in front of you

A \$9.5 B host market — and a \$1.5 B niche moving with it.

Pharma analytical testing vs. CCIT carve-out, 2024 — 2030



Pharma analytical testing outsourcing: \$9.5 B (2025), ~9% CAGR.

CCIT services carve-out: \$1.5 B (2025) → \$2.4 B (2030).

BioPlan 2025: outsourcing budgets surged 11%, U.S. captured 75% of intentions. Top-5 CDMOs hold only 15%.

Sources: Precedence Research, Grand View, MarketsAndMarkets, BioPlan Associates 22nd Annual Report (Jul 2025). Composite midpoints.

Five forces, one quarter

Each alone would justify a faster plan.
All five landed in the same window.

USP <382>

Dec 1 2025. System-level elastomer testing. Burden shifts to drug manufacturer. USP: 'early adoption encouraged.'

Annex 1

EU GMP Annex 1 requires validated deterministic CCIT. Grade A bioContamination limit = zero. Lyo pre-batch sterilization enforced.

GLP-1 / PFS

Prefilled syringes \$9.7B → \$18.1B by 2031. GLP-1 autoinjectors at 15.6% CAGR. COP barrels adopted; methods reset.

503B sterility crisis

GenoGenix recall Jul 2025. ProRx Oct 2025: 36,000+ semaglutide vials. 1,150 adverse events filed YTD. 503B is now a CCIT buyer.

Biologics share

~40% of FDA novel approvals. Every one in a sterile primary container. Glass-delamination recalls back in 2026 headlines.

New containers. New methods. No incumbent.

Each new format is a discrete CCIT method-development engagement.

The 2024 — 2026 wave: container formats, regulation, and CS Analytical's moves



None of the generalists has these validated.

CS Analytical's CSO has publicly presented on cryogenic CCIT — the only credentialed claim.

Every launch above is a billable engagement waiting to be sized.

December 1, 2025. Two things became official.

The regulator is no longer agentically naive.

USP <382>

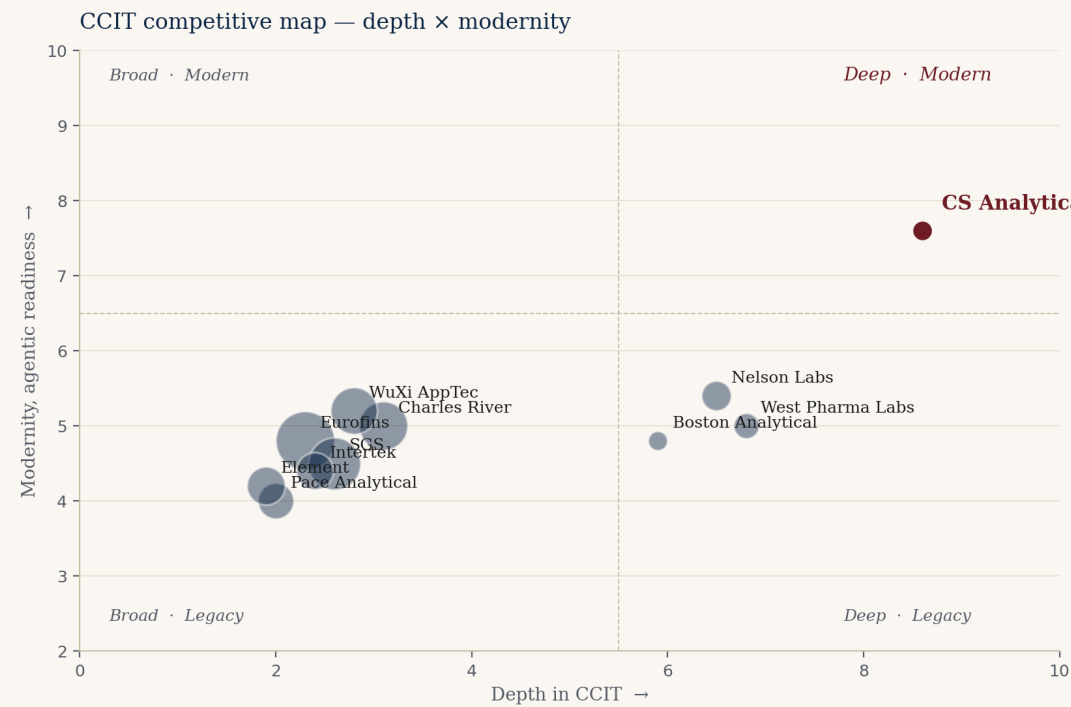
Elastomeric Component Functional Suitability — official.
System-level testing. Responsibility shifts from elastomer
supplier to drug manufacturer. USP: 'early adoption is encouraged.'

The FDA — same day

Agentic AI deployed to all FDA employees. Secure GovCloud, no
training on industry submissions. Supports pre-market review,
post-market surveillance, INSPECTIONS, and compliance.

Who else is in the room

Generalists are broad.
Specialists are scaling. The corner is open.



Three competitive moves in the last six months:

Nelson Labs doubling cleanrooms and publicly acquisitive (Sotera Q1).

West Pharma 165k sq ft Dublin (Mar 2026) for GLP-1 vertical integration.

Eurofins adding 157k m² of lab space in 2025-2026.

What CS Analytical is already doing right

This is not a turnaround brief.
It is an acceleration brief.

RM Analytical · Feb 2026

Raw material and excipient testing under a sister brand.

Clifton expansion · May 2026

Doubling the lab footprint. Capacity ahead of demand.

Micro, gas, IV bag

USP <87>, <788>, gas. IV bag CCIT in 2026 catalog.

Interphex 2026 showcase

IV bag, USP <382>, cell-therapy distribution testing.

Five paths that don't require an AI partner

Each is buildable on the
existing team and footprint.

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- | | | |
|----|-------------------------------------|--|
| 01 | USP <382> wave capture | <i>Named lab for the post-Dec 2025 elastomer rush.</i> |
| 02 | GLP-1 / PFS HVLD productization | <i>HVLD for COP-barrel autoinjectors. High margin.</i> |
| 03 | Annex 1 European mandate | <i>EU GMP Annex 1 opens European sponsors.</i> |
| 04 | Cell, gene & cryo container methods | <i>No incumbent method. CSO credentialed for this.</i> |
| 05 | Regulatory consulting carve-out | <i>Productize the advisory work given away today.</i> |

Where TSI-Citadel multiplies what's already here

McKinsey: 35-45% productivity gain across 270 pharma workflows.

Five agents we would build first.

USP <382> wave-capture

Monitors every 483 / WL for elastomer findings. Resolves to sponsor list. Drafts targeted outreach.

Container-format radar

Tracks SCHOTT, Stevanato, BD, West, Bonfiglioli launches. Sizes the method-dev opportunity.

Inspection-readiness mirror

Mirror to FDA's own agentic inspector. Predicts what AI-augmented inspectors will find.

CAR-T cryo method pipeline

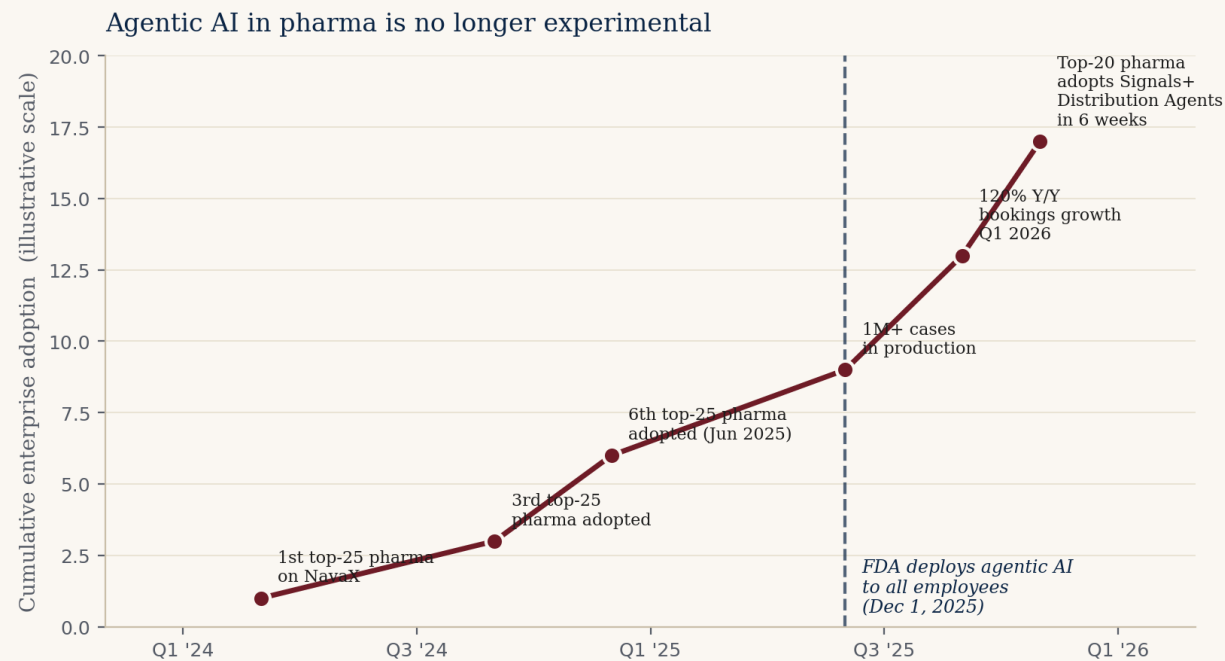
Phase II/III CAR-T radar. Cryo container choices. Failure-rate economics (3.87%-25%) drive demand.

Strategic-optionality tracker

CFO-only quarterly view. Comparable transactions, acquirer activity, market position. Board-friendly.

Six top-25 pharmas. One million cases. 95% accuracy.

Agentic AI in pharma is no longer experimental.



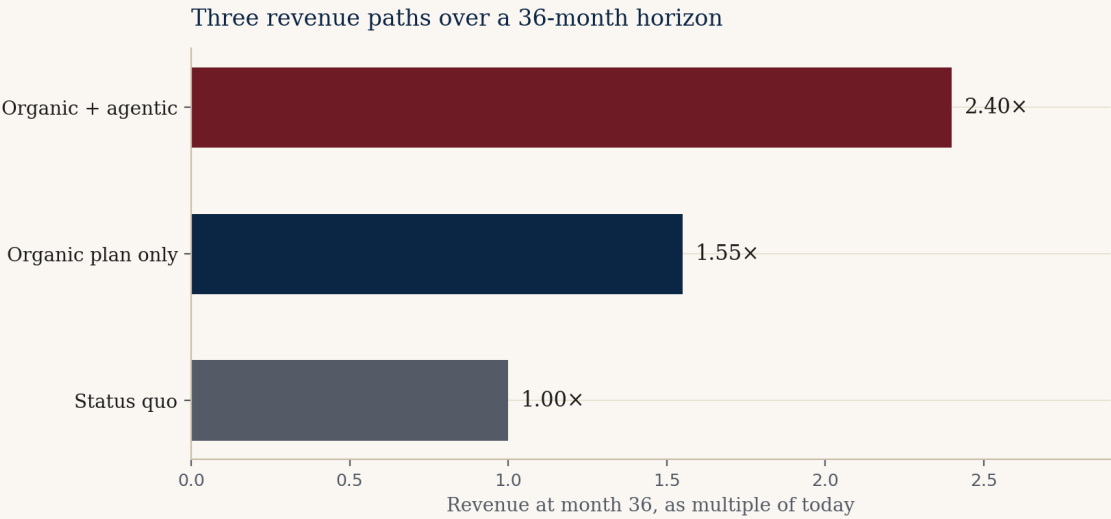
ArisGlobal NavaX: 1M+ cases. 95% accuracy. 30% efficiency. 120% Y/Y bookings Q1 2026.

Sixth top-25 pharma by Jun 2025. Top-20 adopted Signals + Distribution Agents in 6 weeks.

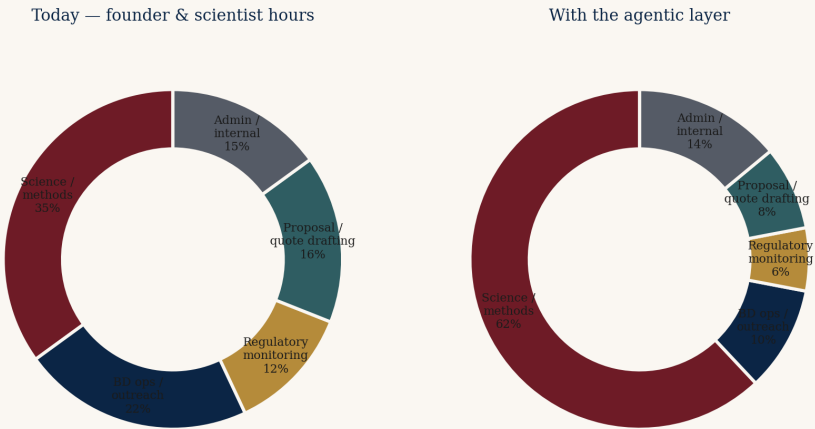
FDA itself deployed agentic AI Dec 1, 2025. The question is no longer if. It is who, when, and at what tempo.

Three revenue paths. One calendar recovery.

Conservative pays for itself.
Aggressive changes the category.



Calendar recovery — where the agentic layer returns time



Strategic optionality: Nelson Labs (Sotera) publicly named pharmaceutical-capability acquisitions a 2026 priority. The 2018 Gibraltar Labs (NJ) precedent is on the record. Life-sciences tools subsector: 18-25x EBITDA on recurring revenue.

Nelson Labs has done this before. In New Jersey.

The Gibraltar precedent.

AUGUST 2018

The transaction

Sotera Health's Nelson Labs acquires Gibraltar Laboratories. Fairfield, NJ. Family-owned since 1970. FDA-registered. ISO 17025. USP-compendial microbiology and analytical chemistry. Two tri-state facilities.

MAY 2026

The statement

Sotera Health Q1 2026 earnings call: 'Targeted acquisitions or partnerships could add testing capability or technology depth.' Management 'exploring strategic acquisitions to enhance Nelson Labs' pharmaceutical capabilities.'

The structural parallel is uncomfortable to ignore.

Prepared for Adam Weiss, CEO, CS Analytical Laboratory · Confidential

Where else this team could play

Adjacencies, not pivots.
Optional, not required.

Combination products	Cross-center (CDER + CDRH) submissions. Few labs ready.
Device sterilization validation	ISO 11135 / 11137 adjacency. Same sponsor base.
Cell-therapy distribution	ISTA/ASTM testing for live-cell packaging. In Interphex slate.
Smart-packaging integrity	Embedded RFID + tamper-evident integrity (TOPPAC infuse class).
Compounding pharmacy	USP <797> / <800>. Fragmented buyer base, less competition.
Veterinary biologics	Same containers. USDA channel. Underpriced relative to human side.

What we'd want to disagree about

Three CFO-relevant risks, three handlings.

Capital timing

Lab expansion + RMA + agentic in one fiscal year. Sequence agentic behind the lab build; agentic pays back in <12 months.

Margin dilution from adjacencies

RMA and micro carry lower margin than CCIT. Hold CCIT pricing while RMA scales; agentic intel keeps the core pipeline full.

Hallucination in regulated context

Human-in-the-loop on every agent flag touching a filed method. Agents are suggestive, not authoritative — by design.

How TSI-Citadel would engage

Light footprint. Boxed scope.
The CFO holds the gate at day 30 and day 90.

DAYS 1 — 30

Diagnostic

Map data feeds, regulatory cycles, inquiry-to-quote flow. Deliverables: exposure map, BD pipeline seed, regulatory-intel trial. No code yet. Go / no-go at day 30.

DAYS 31 — 60

Two agents live

Regulatory-intel and market-radar agents wired to real feeds. Daily output to a named internal user. First measurable lift inside the period.

DAYS 61 — 90

Decide

Joint readout. Numbers. The three remaining agents built only if the first two earned them. Fixed pricing at outset. CFO holds the gate.

THE NEXT CONVERSATION

Read the long form
on a plane.
Then tell us where
we're wrong.

Bruce Longley

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